

Quadrant Technologies

GlobalTextiles Inc.

Statement of Work

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1. Executive Summary

GlobalTextiles Inc., a leader in global textile manufacturing, aims to modernize its enterprise data platform to support robust analytics and reporting across its diverse operations. Due to the rapid growth and geographic expansion, GlobalTextiles faces increasing challenges stemming from siloed data residing in ERP, MES, and supply chain systems. Consequently, leadership lacks a unified, timely, and accurate view of production efficiency, supply chain optimization, and key financial metrics, impeding data-driven decision-making. Quadrant Technologies, in partnership with DataWiz Solutions, proposes implementing a modern medallion data architecture using Microsoft Fabric, supported by enterprise-grade security via Entra and seamless Power BI integration. This initiative will facilitate the consolidation of disparate data sources into a scalable data lake. The architecture will bring standardized processes for data ingestion, transformation, governance, and analytics, specifically tailored to deliver actionable insights for production, supply chain, and finance teams. The project will deliver a purpose-built analytics platform incorporating automated daily data updates, comprehensive data lineage and traceability, and automated quality validation. By leveraging batch ETL processes and proper role-based access controls, GlobalTextiles will be assured of compliance with GDPR and industry standards while enabling governed self-service analytics for business users.

2. Objective

The core objective for this engagement is to design, implement, and operationalize a modern analytics platform for GlobalTextiles Inc. using Microsoft Fabric and Power BI. The solution will unify data from multiple business-critical systems—ERP (SQL Server), MES (CSV), and supply chain partners (REST APIs)—into a single, governed analytics environment. This will empower business teams to drive performance improvements through enhanced visibility, reliable reporting, and operational dashboards.

Specific objectives include:

- Implement the medallion architecture (bronze, silver, gold layers) for high-quality, traceable analytics data.
- Establish secure, role-based access and ensure full compliance with relevant data privacy regulations, including GDPR.
- Enable automated daily data ingestion, transformation, and quality validation with data lineage tracking.
- Deliver self-service Power BI dashboards for production efficiency, supply chain optimization, and financial reporting.
- Facilitate knowledge transfer, documentation, and provide post-go-live support to ensure platform sustainability.

Out-of-scope requirements for this phase include real-time event streaming, integration with legacy mainframe systems, advanced predictive analytics, and non-Microsoft environments.

3. Proposed Solution

Below is a schematic representation of the proposed Medallion Architecture:

The solution centers around Microsoft Fabric, configured to ingest data from ERP (SQL Server), MES (CSV), and supply chain partners (REST APIs). Ingestion is handled through Fabric Data Factory, orchestrating scheduled batch loads and leveraging modular reusable pipelines for future extensibility. Data lands in the Bronze Layer (raw, untouched), transitions through the Silver Layer where cleaning, standardization, and basic validation (including schema harmonization and manipulation of nulls/duplicates) occurs, and finally reaches the Gold Layer as business-ready entities tailored for analytics use cases—production efficiency, supply chain metrics, and financial reporting.

Data transformation and validation leverage Fabric Notebooks for automated data quality checks, audit reports, and lineage tracking. Governance and compliance are assured with Microsoft Entra integration, enforcing role-based access control and row-level security. Encryption is designed for data at rest (in Fabric Data Lake) and in transit (SSL/TLS), directly supporting GDPR obligations and future regulatory initiatives. End-to-end monitoring, error handling, and notification alerts are built into all scheduled processes.

Power BI is natively integrated atop the Gold Layer, enabling business users to build, refresh, and share dashboards on-demand with best-practice templates and visualizations. Metadata management (including data dictionary and data lineage visualization) further empowers both business and technical users. Out-of-scope functionalities for this phase include real-time streaming and cross-cloud integrations; these items can be evaluated for subsequent project phases.

4. Timeline and Resource Details

S No	Activities	Wk1	Wk2	Wk3	Wk4
1	Project Kickoff & Requirements Workshops	X			
2	Source System Analysis & Connectivity Setup	X	X		
3	Fabric Data Lake & Data Factory Configuration		X	X	
4	ETL Pipeline Development (Bronze, Silver, Gold)		X	X	
5	Automated Data Quality Checks & Lineage Tracking			X	

S No	Activities	Wk1	Wk2	Wk3	Wk4
6	Security Layer Implementation (RBAC, Encryption)			X	
7	Power BI Dashboard Development & Integration			X	X
8	User Acceptance Testing & Validation				X
9	Documentation & Knowledge Transfer				X
10	Go-Live & Post-Go-Live Support				X

S No	Role	Expertise	Location
1	Technical Architect + Project Manager	Data Platform Design, Project Governance (Fabric, Security)	Offshore
2	Sr Data Engineer	ETL, Data Lake, Pipeline Automation (Fabric, Data Factory)	Offshore
3	Data Engineer	Pipeline Development, Quality Checks (Fabric, Notebooks)	Offshore
4	Sr Power BI Developer	Dashboard Architecture, UX (Power BI Integration)	Offshore
5	Data Quality Analyst	Test Planning, Reconciliation, Compliance Audit	Offshore
6	Security Specialist	Azure AD/Entra, RBAC, Encryption Setup	Offshore
7	Documentation & Training Lead	Technical Writing, Workshop Facilitation	Offshore

Project Duration: 10 weeks

Cost Breakdown:

- Phase 1 (Weeks 1-4): Data Connectivity, Lake Architecture, ETL Pipeline Build – \$32,000
- Phase 2 (Weeks 5-8): Dashboarding, Security Layer, QA/UAT – \$23,600
- Phase 3 (Weeks 9-10): Documentation, Knowledge Transfer, Go-Live Support – \$10,000
- **Total Services Cost:** \$65,600 (Microsoft Funded: \$15,360, Client to cover remainder unless otherwise funded)

Payment Terms: 50% on kickoff, 30% on UAT completion, 20% post go-live.

5. Deliverables

The scope of deliverables for GlobalTextiles Inc. includes:

- Data Lakehouse Foundation: Fabric Data Lake and Data Factory setup, with all key data sources onboarded (ERP, MES, Supply Chain APIs) and daily batch schedule configured.
- Medallion ETL Pipelines: Automated modular pipelines implementing bronze (raw), silver (cleansed/validated), and gold (business-ready, analytics-optimized) layers, including data quality validation and lineage tracking.
- Security & Governance: Role-based access and encryption implemented, with access logs and compliance documentation per GDPR requirements.
- Power BI Analytics: Three interactive dashboards (Production Efficiency, Supply Chain Health, Financial Reporting) with self-service data access and documentation for business users.
- Testing & Quality Assurance: User Acceptance Test plan, QA test results, sample reconciliation reports, and built-in monitoring/notifications for pipeline health.
- Documentation & Training: Platform architecture document, data dictionary/lineage report, end-user guide, and delivery of one knowledge transfer workshop to client SMEs.
- Post-Go-Live Support: One week of hypercare (ad hoc Q&A, issue resolution, and knowledge reinforcement post-handover).

All deliverables will be associated with acceptance criteria and reviewed by nominated client stakeholders prior to sign-off.

Non-functional requirements include system performance (supporting daily 500GB batch loads within a 2-hour window), system availability (99.5% uptime for analytics), security (encryption, RBAC, GDPR compliance), and maintainability (modular, documented ETL pipelines).

The following are strictly out-of-scope: integration with legacy mainframe, real-time streaming/IoT, machine learning/modeling, and non-Microsoft BI tools.

6. RACI Matrix

Client Responsibilities

- Provision timely access to all required source systems (ERP, MES, supply chain APIs) and deliver database credentials.
- Nominate a business owner and technical SME for requirements workshops, deliverable review, and sign-off.
- Share sample datasets, business logic documentation, and reporting templates relevant to analytics use cases.
- Ensure availability of end users for UAT and provide feedback on dashboard functionality and usability within agreed periods.

- Manage organizational data privacy and GDPR compliance, including continuous oversight of access privileges during the project.
- Coordinate stakeholder participation for knowledge transfer and training sessions.

Vendor Responsibilities

- Design and deliver the entire medallion data architecture using Microsoft Fabric aligning with business, functional, and non-functional requirements.
- Develop, validate, and document ETL pipelines for all three layers with comprehensive lineage and quality checks.
- Implement robust security and governance, including RBAC, encryption, and GDPR-ready processes.
- Configure Power BI dashboards and train business users on report usage, filters, and self-service analytics.
- Conduct quality assurance activities, including reconciliation, pipeline monitoring, and defect management.
- Facilitate recurring project status meetings, progress reporting, and risk escalations as per agreed cadence.
- Prepare all necessary documentation and lead handover/knowledge transfer sessions for client enablement.

7. Assumptions

- GlobalTextiles Inc. will provide access to all necessary source systems and required credentials prior to ETL development kickoff.
- Microsoft Fabric environment, licensing, and network access will be provisioned by the client, with Quadrant Technologies providing guidance for sizing and configuration.
- ERP, MES, and supply chain data is accessible and can be refreshed daily; schema structure will remain stable for the duration of the project.
- The initial scope focuses on batch data ingestion, transformation, and reporting; changes to use-case scope or additional integrations will require a formal change process.
- Client stakeholders will be available for acceptance review, UAT, and knowledge transfer within agreed timelines.
- Power BI licenses for dashboard consumers will be managed by GlobalTextiles according to the user base size specified for this phase.
- The solution will operate fully within the GlobalTextiles Microsoft Azure tenant; no cross-cloud or hybrid deployments are included in this statement of work.

- All compliance with data privacy regulations beyond GDPR (e.g., CCPA, other regional penalties) is managed by the client.

8. Infrastructure and Licensing Requirements

1. Microsoft Fabric

SKU	Capacity Unit (CU)	Pay-as-you-go	Reservation (1 Year)	Benefits
F16	16	\$2,102/month	\$1,251/month	~41% savings
F32	32	\$4,204/month	\$2,502/month	~41% savings
F64	64	\$8,409/month	\$5,002/month	~41% savings

For initial deployment and data volumes (~500GB, batch), F32 Reserved is recommended. Capacity can be increased as data volumes and use cases expand in future phases.

2. Power BI

Premium Per User Licenses: USD \$24/month/user (recommended allocation: 30 business users plus 5 developers for report building and sharing)

3. Annual Running Cost

Component	Cost
F32 (Reserved)	\$30,024 / Annum
Power BI Premium (30 Users)	\$8,640 / Annum

Additional costs may be incurred if F64 or larger SKUs are required due to data growth or advanced analytics scenarios; these will be evaluated in future phases.

9. Roles and Responsibilities

Role	Responsibilities
Technical Architect	Designs solution architecture, oversees compliance and security standards, leads technical workshops and reporting cadence every Monday.
Project Manager (PM)	Owns project delivery, timeline management, resource

Role	Responsibilities
	allocation, and is the escalation point for delivery issues and scope changes.
Data Engineer	Develops ETL pipelines, manages data connectivity and transformation, implements lineage and automated quality checks.
Power BI Developer	Builds Power BI dashboards, configures report visuals, ensures business logic alignment, and conducts dashboard training sessions.
Data Quality Analyst	Designs and executes test plans, reconciles analytics outputs, and audits for compliance.
Security Specialist	Implements role-based access, configures Entra/Azure AD, oversees encryption setup, and ensures GDPR alignment.
Documentation & Training Lead	Prepares user guides, technical documentation, and facilitates knowledge transfer sessions and go-live checklist review.
Business Users (Client)	Perform UAT, provide feedback and requirements validation, and serve as dashboard content owners post-implementation.

Reporting Cadence: Weekly status reviews (Monday), bi-weekly steering meetings (Thursday), and escalation channel via PM (response within 1 business day).

10. Risks

Risk Register:

Risk	Description	Mitigation	Owner
Data Inconsistency	Source data formats/quality differences between ERP, MES, and supply chain APIs	Automated quality validation, business rule harmonization, and reconciliation reporting	Data Engineer / Data Quality Analyst
Access Delays	Slow access provisioning to source systems	Early request of credentials, escalation procedure to PM if delays > 3 days	Project Manager
Scope Creep	Unplanned use-case additions during build	Strict scope adherence, formal change request process	Project Manager
Security	GDPR or organizational	Regular compliance	Security

Risk	Description	Mitigation	Owner
Compliance	policy breaches	checks, legal review, RBAC audit	Specialist
Performance Bottleneck	Large batch loads impacting nightly ETL window	Pilot load testing, incremental loading optimization, Fabric capacity review	Technical Architect
Stakeholder Availability	Limited end-user input or UAT participation	Engagement schedule, proactive communication, backup reviewers assigned	Project Manager
Integration Failure	Connection or API changes mid-project	Dev/test APIs separately; rollback plan for critical pipeline failures	Data Engineer

Quality Assurance Plan:

- Test ETL pipelines against 3 months of historical data during development; check accuracy and completeness with sample reports.
- Automate data quality checks (nulls, duplicates, outliers) and log exceptions for review on a nightly basis.
- Conduct User Acceptance Testing (UAT) with nominated business owners prior to go-live; all defects tracked in a project issue register.
- Reconciliation reports produced for every dashboard dimension/measure, signed off by data quality analyst.
- Implement monitoring and alerts for ETL pipeline failures, performance anomalies, or security audit failures with a response plan.

11. Success Criteria

- Successful, accurate, and timely daily ingestion of ERP, MES, and supply chain data into Fabric Data Lake (Bronze Layer).
- Completion of ETL development with automated validation and creation of silver and gold layers tailored for analytics use cases.
- Delivery of three Power BI dashboards—production efficiency, supply chain monitoring, financial reporting—with user acceptance meeting or exceeding agreed business requirements.
- Role-based access controls and encryption implemented as specified; compliance with all GDPR requirements validated through audit reports.

- Project documentation, data dictionary, and lineage visuals delivered and signed off by client SMEs.
- Successful completion of knowledge transfer session with positive client feedback.
- All agreed success metrics for data quality (accuracy $\geq 98\%$), refresh timeliness (batch < 2 hours), and system availability met during post-go-live support period.
- No unresolved critical or major defects at project closure.

12. Commercial

Item	Details
SOW Effective Date	July 8, 2024
SOW Expiration Date	September 16, 2024
Duration	10 Weeks
Total Cost	\$65,600 (Microsoft Funded: \$15,360)

Payment Terms: 50% due at project kickoff, 30% on completion of UAT and dashboard delivery, 20% post go-live and acceptance. All payments are due NET 30 days from invoice, payable in USD. Taxes are additional as applicable.

Client

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